Python Assignment 3 Documentation

Topic: Loops, Control Statements & Functions

Overview

This document provides the structural breakdown and functional documentation for the executable Python script solving Assignment 3. The assignment highlights the implementation of loops, conditional structures, and modular function programming.

1. Number Guessing Game (While Loop & Control Statements)

Logic & Implementation: Implements a numeric game using a while loop capped at 3 valid structural attempts. It processes user input dynamically while actively applying loop controls:

• continue: Automatically shifts iteration to bypass standard verification check sequences if input numbers are outside the designated bounds (1 to 10).

• break: Instantly terminates execution flow upon a correct matching condition evaluation.

• else structural clause: Triggers conditionally only if the loop pool finishes all planned iteration limits without executing a break intercept routine.

2. Multiplication Table Generator (For Loop)

Logic & Implementation: Leverages a strict deterministic for loop mapped explicitly across a range(1, 11). It processes an algebraic input multiplication index and cleanly logs data directly matching standardized presentation rules.

3. Body Mass Index Calculator (Functions)

Logic & Implementation: Encapsulates logic processing computations into a dedicated named structural definition block: calculate_bmi(weight, height). It dynamically executes input values and processes float calculations cleanly matching the standardized index matrix.

Script File Architecture Reference

Component / Routine	Target Element Details
number_guessing_game()	Tracks state tracking limits, attempts, range validation checks, and targets.
multiplication_table_generator()	Drives numeric processing loop mapping indexes up to factor 10.

calculate_bmi(w, h)	Mathematical execution handler: Returns float values based on mass index formula.

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